Product Requirement Document (PRD)

Himalaya Carbon Tokenization Engine

Registry-Synchronized Carbon Market Infrastructure for Bhutan

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1. Product Overview

The **Himalaya Carbon Tokenization Engine** is a digital infrastructure platform designed to improve the **trading, transparency, and retirement lifecycle of Bhutan's officially issued carbon credits**.

The platform does not create carbon credits.
Instead, it acts as a **secondary market layer that tokenizes registry-backed carbon units**, enabling easier trading, fractional exposure, and verifiable retirement.

The system integrates with Bhutan's **National Carbon Registry, CAD Trust metadata network, and Article 6 international transfer framework**, ensuring that all tokenized assets remain legally synchronized with sovereign registry records.

The primary goal is to solve market inefficiencies in the global voluntary carbon market by introducing a **transparent, registry-linked trading and retirement interface**.

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2. Product Vision

To create the **world's first sovereign-aligned carbon liquidity layer**, enabling transparent and auditable digital trading of registry-backed carbon units without compromising environmental integrity or accounting rules.

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3. Problem Statement

The global voluntary carbon market suffers from several structural weaknesses:

Market Fragmentation

Carbon credit data is dispersed across multiple registries, brokers, and private contracts, making verification difficult.

Lack of Transparency

Price discovery is weak because most carbon credit transactions occur privately.

Liquidity Constraints

High-quality credits can be difficult to purchase in small or flexible quantities.

Double-Counting Risk

Improper tokenization can result in the same credit being used in multiple markets simultaneously.

These inefficiencies reduce trust and limit participation in carbon markets.

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4. Product Solution

Himalaya introduces a **registry-synchronized tokenization layer** that enables digital trading while preserving registry authority.

The system operates using a **multi-layer architecture**.

Layer 1 — National Carbon Registry

The Bhutan National Carbon Registry remains the **legal system of record**.
It manages issuance, authorization, transfer, cancellation, and retirement of carbon units.

Layer 2 — CAD Trust Metadata Layer

CAD Trust provides standardized metadata and transparency across registries to prevent double counting and improve auditability.

Layer 3 — Himalaya Tokenization Layer

The Himalaya engine tokenizes registry-backed units after they are **immobilized in the registry**, ensuring that each token corresponds to a single verified carbon unit.

Layer 4 — On-Chain Market Infrastructure

Tokenized units can be traded, pooled, or retired through blockchain-based market tools.

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5. Target Users

Corporate Carbon Buyers

Companies purchasing credits for sustainability commitments.

Financial Institutions

Institutions seeking exposure to high-integrity carbon assets.

Climate Funds

Funds investing in nature-based carbon projects.

Retail Climate Participants

Individuals wishing to purchase and retire carbon units.

Government Agencies

Regulators and policymakers monitoring carbon credit flows.

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6. Core Product Features

6.1 Registry-Synchronized Tokenization

Carbon tokens are minted only when corresponding units are immobilized in the national registry.

Each token is linked to:

- project ID
- vintage year
- methodology
- authorization status
- registry unit ID

6.2 Canonical Carbon Asset Layer (ERC-1155)

The system uses **ERC-1155 multi-token standard** to represent multiple carbon asset categories in a single contract.

Each token class corresponds to a unique carbon credit category.

Example token class:

Bhutan Forest Conservation Project
Vintage 2025
Article 6 Authorized

6.3 Liquidity Layer (ERC-20 Wrappers)

To support decentralized exchanges and automated market makers, ERC-1155 tokens can be wrapped into **ERC-20 trading tokens**.

This enables:

- liquidity pools
- decentralized trading
- price discovery

6.4 Carbon Retirement Dashboard

A user interface allowing buyers to retire credits.

When a token is retired:

1. the token is burned on chain
2. the registry records cancellation
3. CAD Trust reflects updated status
4. the user receives a verifiable certificate

6.5 Fractional Exposure

Users can trade fractional token shares on-chain, while final retirement occurs in **whole registry units**.

This enables:

- small purchases
- aggregated retirement events
- improved liquidity

6.6 Metadata Transparency

Each token includes publicly verifiable metadata linked to CAD Trust and the national registry.

Displayed information includes:

- project documentation
- credit vintage
- verification status
- authorization status
- retirement history

6.7 Compliance and Governance Controls

Administrative controls enable regulators or operators to:

- pause token transfers
- freeze assets
- handle disputes
- revoke tokens if registry status changes

These controls ensure alignment with international carbon accounting frameworks.

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7. System Architecture

Off-Chain Components

Registry Integration Service
Synchronizes token state with registry status.

Metadata Service
Fetches project information through CAD Trust APIs.

Oracle System
Attests registry state to smart contracts.

Compliance Engine
Manages KYC/AML requirements.

On-Chain Components

ERC-1155 Token Contract
Represents registry-backed carbon units.

ERC-20 Wrapper Contracts
Enables liquidity pool trading.

Retirement Contract
Handles token burn and retirement verification.

Governance Contract
Provides emergency administrative controls.

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8. MVP Scope

The minimum viable product should focus on **core infrastructure rather than full DeFi functionality**.

MVP Features

- permissioned minting module
- registry synchronization service
- CAD Trust metadata display
- ERC-1155 token system
- retirement dashboard
- compliance administration panel

Advanced features such as AMM trading pools can be introduced later.

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9. Regulatory Alignment

The platform must align with Bhutan's **Carbon Market Rules 2023** and Article 6 requirements.

Key regulatory considerations include:

- registry authority remains primary
- corresponding adjustment transparency
- authorization tracking
- compliance with national climate policy

Deployment may occur through Bhutan's **FinTech regulatory sandbox**.

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10. Revenue Model

Tokenization Fees

Charged when registry units are converted into tokenized assets.

Trading Fees

Collected on token swaps or liquidity pool transactions.

Enterprise Services

Subscription fees for corporate retirement dashboards and reporting tools.

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11. Key Risks

Governance Risk

Lack of official recognition could undermine credibility.

Registry Synchronization Risk

Token supply must always equal immobilized registry units.

Regulatory Risk

Tokenization must align with registry rules and international standards.

Liquidity Risk

Market adoption depends on sufficient trading activity.

Oracle Integrity Risk

Incorrect registry data could compromise token validity.

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12. Success Metrics

Key indicators of platform success include:

- number of tokenized carbon units
- trading volume
- retirement transactions
- institutional adoption
- integration with international carbon markets

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13. Implementation Roadmap

Phase 1

Registry integration and tokenization infrastructure

Phase 2

Market interface and trading capabilities

Phase 3

Corporate retirement tools and reporting systems

Phase 4

International market integrations

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14. Strategic Positioning

Himalaya should be presented as:

Bhutan's registry-synchronized carbon market infrastructure

not as

a new carbon registry or independent token issuance platform.

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15. Executive Summary

The Himalaya Carbon Tokenization Engine is a **secondary-market infrastructure layer designed to improve trading, transparency, and retirement of Bhutan's officially issued carbon credits**.

By synchronizing blockchain tokens with the national carbon registry and CAD Trust metadata network, the system enables digital market participation without compromising environmental accounting integrity.

If implemented correctly, the platform could position Bhutan as a global leader in **high-integrity carbon market infrastructure**.

A deeper step is possible from here. The document above is a **product PRD**, but the system actually sits at the intersection of **climate policy, blockchain architecture, and sovereign market infrastructure**.